EV MARKET ANALYSIS DASHBOARD

A Data Visualization Project Using Tableau Public 2026.3

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PROJECT REPORT

Abstract

This project presents an interactive dashboard for exploring electric vehicle (EV) registration data using Tableau Public 2026.3. The dashboard combines four views: the top EV manufacturers by distinct registered vehicle IDs, vehicle-type distribution, registrations by model year, and geographic distribution. The supplied CSV dataset includes fields such as DOL Vehicle ID, Make, Model Year, Electric Vehicle Type, County, City, State, and Postal Code. Distinct counting of vehicle IDs is used in the visible charts to reduce the risk of counting repeated IDs as separate vehicles. The dashboard is intended to make category comparisons and broad patterns easier to inspect. Values reported in this document are limited to those visible in the current dashboard screenshots; other numerical results are marked for verification rather than inferred. The model-year view was filtered to 1999–2025 during review, and this cutoff should be retained only after confirming that later-year records are incomplete or otherwise outside the intended analysis.

Keywords: Electric vehicles, Tableau Public, data visualization, registrations, dashboard, exploratory data analysis

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1. Introduction

Electric vehicles are an important part of the changing transportation landscape. Registration datasets can help users explore which manufacturers have the most registered vehicles, how registrations are distributed across vehicle types, how counts vary by model year, and where registered vehicles are geographically concentrated. However, large tables are often difficult to interpret quickly. Data visualization provides a way to summarize the data and communicate patterns in a compact, accessible format.

This project uses Tableau Public to build an EV Market Analysis Dashboard from the supplied export.csv dataset. The final dashboard contains four coordinated visualizations and is saved as a Tableau packaged workbook (.twbx).

2. Problem Statement

The raw EV registration dataset contains multiple fields and a large number of records. Reading the data row by row does not provide an immediate overview of manufacturer rankings, vehicle-type composition, model-year patterns, or geographic concentration. The project addresses this problem by transforming selected fields into clear charts that support exploratory analysis.

3. Objectives

- Compare manufacturers by the number of distinct registered vehicle IDs.
- Compare the registration counts of the vehicle types represented in the dataset.
- Visualize the relationship between model year and distinct registered vehicle IDs.
- Display geographic patterns using the available county and state fields.
- Combine the visualizations into a single dashboard for quick review.
- Validate workbook reopening and data connectivity before submission.

4. Scope of the Project

The project is an exploratory visualization of the supplied dataset. It describes patterns present in the available – records and does not establish causal relationships, forecast future adoption, or represent the entire electric vehicle market unless the source dataset's coverage supports such claims. The geographic visualization appears to focus on Washington State counties; this scope should be described accordingly unless the data source is verified to cover a wider area.

5. Dataset Description

The input file is named export.csv. The fields visible in Tableau's Data pane include the following:

Field	Purpose in analysis
DOL Vehicle ID	Vehicle identifier; used with a distinct-count aggregation in the displayed charts.
Make	Manufacturer category for the top-manufacturer chart.
Model Year	Year dimension for the registration trend.

Electric Vehicle Type	Vehicle-type category used to compare BEV and plug-in hybrid registrations.
County / City / State	Geographic dimensions for mapping and regional analysis.
Postal Code	Additional location field available in the source data.
Model	Vehicle model information available for possible detailed analysis.

Dataset size, source URL, extraction date, exact row count, missing-value count, and duplicate-record count were not confirmed in the screenshots available for this report. These details should be added only after checking the original dataset or its documentation.

6. Tools and Technologies

- Tableau Public 2026.3: used to connect to the CSV, create worksheets, and assemble the dashboard.
- CSV (Comma-Separated Values): tabular source format for the registration dataset.
- Tableau packaged workbook (.twbx): final deliverable format that packages workbook content with local data where supported.
- Windows Snipping Tool: optional tool for capturing a screenshot of the completed dashboard.

7. Methodology

1. Connect the export.csv file to Tableau Public and inspect the available fields.
2. Check field types and identify the dimensions and measures required for each visualization.
3. Create a manufacturer comparison using Make and distinct count of DOL Vehicle ID.
4. Create a vehicle-type comparison using Electric Vehicle Type and distinct count of DOL Vehicle ID.
5. Create a line chart using Model Year and distinct count of DOL Vehicle ID.
6. Create a geographic visualization using the available geographic fields and appropriate geographic roles.
7. Arrange the four worksheets on one dashboard, add the title, and adjust the dashboard size and chart layout.
8. Review filters and aggregation choices, save the workbook as a packaged workbook, close it, and reopen it to test that the file loads.

8. Dashboard Design and Visualizations

8.1 Top 10 EV Manufacturers by Registered Vehicles

This horizontal bar chart ranks manufacturers using distinct count of DOL Vehicle ID. The current dashboard screenshot visibly shows Tesla as the longest bar. The screenshot displays approximately 122,291 for Tesla. This value should be treated as a screenshot-observed value and rechecked in Tableau before being quoted as a final result. Other visible labels include Ford (15,997), Chevrolet (20,218), Audi (6,053), Hyundai (12,581), and Porsche (2,170); the exact ordering and labels should be verified because the screenshot resolution and bar labels may not show every value clearly.

8.2 Vehicle Distribution by Type

This horizontal bar chart compares the categories shown in Electric Vehicle Type. The screenshot visibly displays Battery Electric Vehicle at approximately 241,840 and Plug-in Hybrid Electric Vehicle at approximately 58,302.

These are values observed in the screenshot and should be confirmed in the worksheet before final reporting. The chart suggests that the battery-electric category has a larger distinct vehicle count in the displayed data.

8.3 EV Registrations by Model Year

This line chart plots Model Year against distinct count of DOL Vehicle ID. The displayed trend rises substantially in more recent model years, with a visible peak around 2023 and lower values afterward in the unfiltered view. During validation, a Model Year range of 1999–2025 was selected to avoid displaying the later-year tail while its completeness is investigated. The exact cutoff should be justified using the source data; it should not be selected solely because it makes the line look smoother.

8.4 EV Registrations by Geographic Area

The map displays geographic areas that appear to be counties in Washington State, shaded according to the selected measure. The darkest areas indicate comparatively higher values in the displayed map. The exact county names, counts, and geographic coverage were not confirmed from the screenshots. Before reporting a specific leading county, verify the map's tooltip and the geographic field configuration.

9. Results and Observations

The following observations are supported by the visible dashboard, subject to rechecking the underlying worksheet values:

- Tesla is the most prominent manufacturer in the displayed manufacturer chart.
- The Battery Electric Vehicle category has a substantially longer bar than the Plug-in Hybrid Electric Vehicle category in the displayed vehicle-type chart.
- The model-year line rises notably in the later years shown, but the latest-year values require a completeness check before interpreting the final segment.
- The geographic map indicates uneven registration counts across the mapped counties, with some areas shaded more darkly than others.

Verified numeric findings to insert after checking Tableau:

Metric	Verified value	Verification source
Total distinct vehicle IDs	260,630	Data Source / worksheet
Leading manufacturer and count	TESLA — 122,291	Top 10 EV Manufacturers tooltip
BEV distinct count	241,840	Vehicle Type Analysis tooltip
PHEV distinct count	58,302	Vehicle Type Analysis tooltip
Highest-count county	King County, WA — 143,976	Geographic map tooltip
Latest complete model year	36,561	Model Year filter and source records

10. Data Validation and Quality Considerations

- Distinct count: the displayed charts use COUNTD(DOL Vehicle ID). Confirm that this aggregation is intended and that DOL Vehicle ID is populated consistently.

- Model-year range: years after 2025 were excluded in the trend during review. Check whether those records are incomplete, future-dated, or valid before retaining the filter in the final submission.
- Geographic roles: verify that County and State are assigned appropriate geographic roles and that the map is not unintentionally limited by a filter.
- Filters: ensure the Model Year filter affects only the Model Year Trend worksheet unless a broader dashboard-level filter is intentionally required.
- Reopening test: the saved .twbx workbook was reopened successfully by the user, confirming that the file loads in Tableau Public 2026.3 at that point in the workflow.
- Visual review: check for truncated labels, excessive blank space, missing legends, and unreadable axis labels before exporting the final screenshot.

11. Limitations

- The report is based on the supplied dataset; its completeness, extraction date, and source documentation were not independently established here.
- A distinct vehicle-ID count represents the records included in the dataset and does not automatically equal all vehicles in service or all market sales.
- Recent model years may be incomplete or may contain future-dated records; interpretation requires a data-quality check.
- The map appears to represent Washington counties, so conclusions should not be generalized to all states or the entire United States without broader data coverage.
- Exact values for several measures, the full row count, missing values, and county-level rankings remain placeholders until verified in Tableau.

12. Conclusion

The EV Market Analysis Dashboard demonstrates how Tableau Public can convert a registration dataset into an accessible set of visual summaries. The dashboard supports comparison of manufacturers, vehicle types, model-year patterns, and geographic variation. The current visual review indicates that Tesla is the most prominent manufacturer in the displayed ranking and that battery electric vehicles exceed plug-in hybrids in the displayed category comparison. These observations should be finalized only after verifying the corresponding values and checking the model-year and geographic filters. The packaged workbook provides a reusable starting point for further analysis and presentation.

13. Future Enhancements

- Add KPI cards for total distinct vehicle IDs, leading manufacturer, and most common vehicle type.
- Add interactive filters for model year, manufacturer, vehicle type, and geographic area where they are analytically appropriate.
- Improve tooltips with clearly formatted counts and relevant context.
- Add a source note, data extraction date, and definition of the registration metric.
- Validate and document missing values, duplicate IDs, and incomplete model years.
- If permitted by the assignment, publish the dashboard to Tableau Public and include the public URL in the report.

14. References

- Tableau Public 2026.3, used to create the dashboard. Product documentation: https://help.tableau.com/
- Source dataset: export.csv, supplied for this project. https://catalog.data.gov/dataset/electric-vehicle-population-data?utm_source=chatgpt.com
- Project workbook: EV_Market_Analysis_Dashboard_Final.twbx.



EV_Market_Analysis_Dashboard_Final.twbx

Appendix A: Final Submission Checklist

- Replace all cover-page placeholders with correct student and faculty details.
- Verify every numeric finding against the live Tableau worksheet or tooltip.
- Confirm that the 1999–2025 model-year filter is justified by data completeness.
- Confirm that the geographic map and its filters represent the intended scope.
- Save and reopen the final .twbx workbook.
- Insert a clear screenshot of the final dashboard if required by the instructor.
- Follow the institution's required font, margins, citation style, and report length if these have been specified.